LogiEdge

Phase 1 Report

Requirements, system architecture and hardware selection

Group 35 | AIML ZG535 - Machine Learning on Edge

BITS Pilani Work Integrated Learning Programmes
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1. Scope and project objective

This report consolidates the design for Components A and B and the sensor interfaces that lead into Component C. Phase 2 separately documents implementation and measured validation for Components C-F. These reports are prepared from the completed project records; no earlier phase-submission date is claimed.

FreightBridge operates 85 refrigerated trucks in the pilot, with potential expansion to 265 vehicles. Manual logs taken every two hours cannot provide a 90-second response to a refrigeration fault. LogiEdge therefore places classification, temperature rules and durable event storage beside the truck sensors [1].

Design requirement [1]	Decision / acceptance condition
Local response within 90 seconds	On-node inference and immediate temperature guard
35-90 minute connectivity gaps	Durable local log; queued uplink with acknowledgements
10 W AI power budget	Proposed Pi 5 option; verify whole-node power on target
Three cargo-state classes	Normal, Warning and Critical; operational rules remain explicit
Fleet deployment and monitoring	Versioned models/reference, cached OTA, PSI and idempotent deployment

Prototype boundary. The implemented truck and operations centre are containers on a Windows 11 laptop through Ubuntu 24.04 / WSL2. Sensor readings are simulated. The Raspberry Pi is the proposed field target; physical sensor integration, ARM64 execution and on-truck power consumption have not been measured.

2. Four deployment constraints

Latency. The assignment allows 90 seconds from a fault signature to local detection. A one-degree-Celsius-per-minute rise makes waiting for a two-hour log unacceptable. Connected cloud inference is not ruled out by latency alone: an illustrative 500 ms network allowance plus one second of compute would fit the budget. Those are planning assumptions, not measured rural tail latency. Network outages are the decisive limitation. Local model execution and a raw-temperature guard avoid making the response dependent on a cellular round trip. Initial feature collection, message handling and storage must count in end-to-end timing.

Bandwidth. At four bytes per numerical value, temperature contributes 1 x 4 = 4 bytes/s; raw three-axis vibration contributes 500 x 3 x 4 = 6,000 bytes/s. Across 86,400 seconds this is 518.7456 decimal MB/day/truck, costing Rs 51.87456 at Rs 0.10/MB. Assuming 100 door records/day of 32 bytes each adds 0.0032 MB. An illustrative edge budget of 1,440 one-KB heartbeats plus 20 one-KB alerts is 1.46 MB/day and Rs 0.146, about 99.72% less numerical payload. Encoding, transport, security and retry overhead are excluded. The implemented 0.5 Hz vibration-RMS simulator is already summarised; it does not generate the raw 500 Hz waveform.

Connectivity. The stated 35-90 minute gaps last 2,100-5,400 seconds, or about 23-60 complete alert budgets. A cloud-only system cannot guarantee new decisions during those gaps. LogiEdge continues local processing and writes events to a SQLite-backed outbox. Delivery is retried when the operations service is reachable. A matching acknowledgement marks successful delivery, and stable event identifiers prevent a retry from becoming a second logical incident. Remote receipt necessarily waits for connectivity even while local protection continues.

Privacy. Processing locally reduces routine export of raw cargo-condition data. The uplink can carry incident summaries and selected operational context instead of the complete sensor history. Local inference alone does not fulfil the access-control requirement: production needs authenticated interfaces, encrypted transport and storage, retention rules, signed updates and physical device controls. The local HTTP demonstration is not evidence of these protections. Model and reference hashes detect accidental changes but do not establish authorisation. A private source repository also does not secure a truck or its backend.

Payload basis	MB/day/truck	Rs/day/truck
Raw numeric data only	518.7456	51.87456
Numeric data + assumed door records	518.7488	51.87488
Illustrative edge summaries	1.4600	0.14600

All costs use assignment rates and decimal units. Fleet-scale transfers are projections, not observed truck measurements.

3. Architecture and sensor interfaces

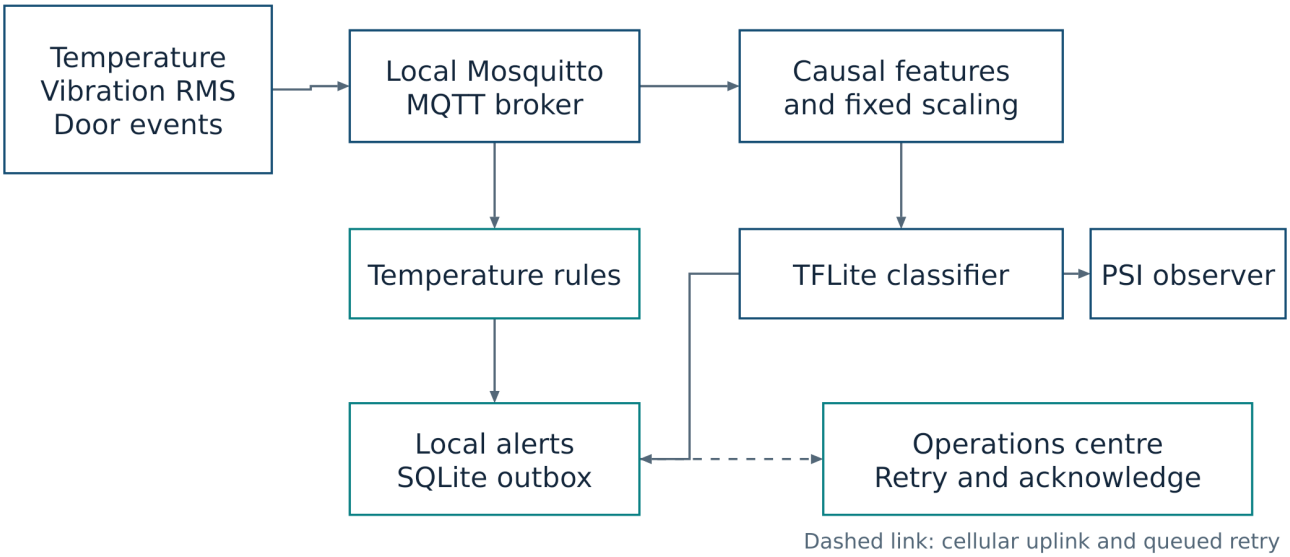


Figure 1. Truck-side sensing, local intelligence and durable uplink. The dashed path represents the proposed cellular connection; the prototype backend is local.

Temperature, vibration RMS and door events reach the local Mosquitto broker. Shared causal preprocessing generates six model inputs. The TFLite classifier publishes its result, while separate temperature rules can raise an alarm before the first complete model window. Local severity accounts for both paths. The operations service acknowledges queued events independently of local decisions.

Topic leaf under logibridge/trucks/{truck_id}/	Purpose
temperature	One-Hz temperature readings
vibration_rms	Half-Hz compressor RMS summaries
door_event	Timestamped OPEN / CLOSE context
inference	Model result, identity, features and operational state
alerts	Changes in local operational severity

All five topics use QoS 1 and non-retained messages. QoS 1 can redeliver, so message identifiers and database uniqueness checks support deduplication; it does not by itself guarantee exactly-once application effects. Heartbeats use the HTTP outbox. Broker health uses logibridge/system/broker_health.

Fusion choice. Sensor-specific filtering precedes concatenation of three temperature and three vibration features. This feature-level interface handles different sampling rates without aligning all raw waveform samples. Door state remains context. Data-level fusion would require additional alignment, while decision-level fusion would require multiple models and reconciliation rules.

4. Hardware selection and Roofline analysis

The Module 2 Constraint Triangle balances **Compute Performance, Power Budget and Unit Cost** [2]. The 10 W budget is the dominant constraint here. The following prices and power figures are assignment assumptions, not current supplier quotations or measured total-system consumption [1].

Option	Rs/truck	Power	85 trucks Rs lakh	265 trucks Rs lakh
Pi 5 8 GB + AI HAT+	15,000	7.5 W	12.75	39.75
Jetson Orin Nano Super	45,000	15 W	38.25	119.25
STM32H7 custom MCU	3,500	0.4 W	2.975	9.275

Proposed target: Pi 5. It fits the assignment power estimate and supports the Linux/container design at lower fleet cost than Jetson. Jetson exceeds the 10 W figure before demonstrating a need for its extra compute. STM32 is cheaper and lower power, and the small model may fit an MCU, but this Python, TensorFlow, Docker and SQLite stack would need a different embedded implementation. The tiny MLP has no demonstrated need for accelerator TOPS.

4.1 Assigned workload calculation

Quantity	Calculation	Result
Arithmetic intensity	45 MFLOP / 18 MB	2.5 FLOP/byte
Ridge point	16 GFLOP/s / 12 GB/s	1.333 FLOP/byte
Compute lower bound	45e6 / 16e9 seconds	2.8125 ms
Memory lower bound	18e6 / 12e9 seconds	1.5000 ms
Roofline classification	Intensity above ridge	Compute bound

For this hypothetical workload, reducing operations or using compatible SIMD / acceleration addresses the compute limit. Quantisation and structured removal still require measurement: smaller files need not execute faster. The assigned 45-MFLOP / 18-MB example is separate from the actual 803-parameter MLP.

Design handoff. Phase 2 checks model accuracy, Critical recall, complete window timing, offline queue recovery, model-cache updates, live PSI and seven-task deployment. Physical Pi latency, working RAM, microSD needs and whole-node power remain target-validation work.

Sources and design records

[1] LogiEdge Mini-Project Problem Statement - Student, AIML ZG535, Components A-F and submission requirements.
[2] Module2_EdgeHardware.pptx, Constraint Triangle and Roofline material. Project design records:

`scenario_architecture/constraint_analysis.md`
`hardware/hardware_justification.md`
`scenario_architecture/system_architecture_diagram.png`